

Experiment 02A

Medical GPT Training Baseline Report

Medical text corpus | GPT-2 BPE / tiktoken | Standard Attention | Lightning H200 | 5 epochs

Report summary

This report documents the scaled GPT-2/tiktoken baseline that followed Experiment 01. Experiment 02A increased dataset scale, context length, model capacity, checkpointing, and training infrastructure to establish a stronger medical-domain baseline.

Dataset	Medical text corpus
Tokenizer	GPT-2 BPE / tiktoken
Model Scale	~194M parameters
Context Length	1024 tokens
Final Perplexity	66.1369
Status	Scaled baseline and transition plan

1. Executive Summary

Experiment 02A is the scaled GPT-2/tiktoken baseline for the medical-domain GPT training pipeline. It was designed as the direct improvement over Experiment 01, which validated the pipeline but suffered from severe overfitting on a small WikiText-103 subset.

In this experiment, the model was trained on medical text chunks using a larger GPT-style decoder-only transformer, a longer 1024-token context window, improved checkpointing, train/validation CSV logging, and a Lightning H200 GPU environment.

Metric	Final Value
Train Loss	3.3203
Validation Loss	4.1917
Validation Perplexity	66.1369
Best Checkpoint	Epoch 5
Best KV-cache Speedup	1.5389x at prompt length 512

2. Experiment Objective

The objective of Experiment 02A was to establish a stronger baseline after the limitations observed in Experiment 01. The tokenizer was intentionally kept as GPT-2 BPE through tiktoken so that the experiment could serve as the generic-tokenizer reference before introducing the custom biomedical BPE tokenizer in Experiment 02B.

Objective	Purpose
Scale the baseline	Move from a small debugging experiment to a more realistic medical-domain training run.
Improve context modeling	Increase the context length from 128 to 1024 tokens.
Improve reproducibility	Save best, latest, epoch checkpoints, configuration, and CSV logs.
Benchmark inference	Measure generation with and without KV-cache.

3. Dataset and Experimental Context

Experiment 02A used medical-domain text chunks prepared from the project corpus. The dataset was used as raw text input and tokenized with the GPT-2/tiktoken tokenizer during the experiment pipeline.

Field	Value
Dataset	Medical text corpus
Input Format	Raw text chunks
Number of Text Chunks	100
Tokenizer	GPT-2 BPE tokenizer through tiktoken
Training Hardware	Lightning AI H200
Experiment Role	Scaled baseline for tokenizer comparison

4. Model Configuration

The model followed a GPT-style decoder-only transformer architecture with standard causal multi-head self-attention. The configuration was significantly larger than Experiment 01 and used a 1024-token context length.

Parameter	Value
Vocabulary Size	50,257
Context Length	1024
Embedding Dimension	768
Attention Heads	12
Transformer Layers	12
Dropout	0.1
QKV Bias	False
FFN Hidden Dimension	3072
Attention Type	Standard causal multi-head attention
Approximate Model Scale	~194M parameters

5. Training Configuration

Hyperparameter	Value
Epochs	5
Batch Size	8
Gradient Accumulation	8 steps
Effective Batch Size	64 sequences
Learning Rate	3e-4
Minimum Learning Rate	1e-5
Learning Rate Schedule	Warmup + cosine decay
Gradient Clipping	1.0
Mixed Precision	Enabled
Checkpointing	best.pt, latest.pt, epoch_N.pt

6. Training Results

Epoch	Train Loss	Validation Loss	Perplexity
1	6.4186	5.1973	180.7757
2	4.5894	4.6690	106.5914
3	3.9408	4.3225	75.3733

4	3.5148	4.2077	67.2047
5	3.3203	4.1917	66.1369

7. Training Curve Analysis

Validation loss decreased from 5.1973 to 4.1917, and validation perplexity decreased from 180.7757 to 66.1369. This was a major improvement over Experiment 01, where validation loss increased throughout training.

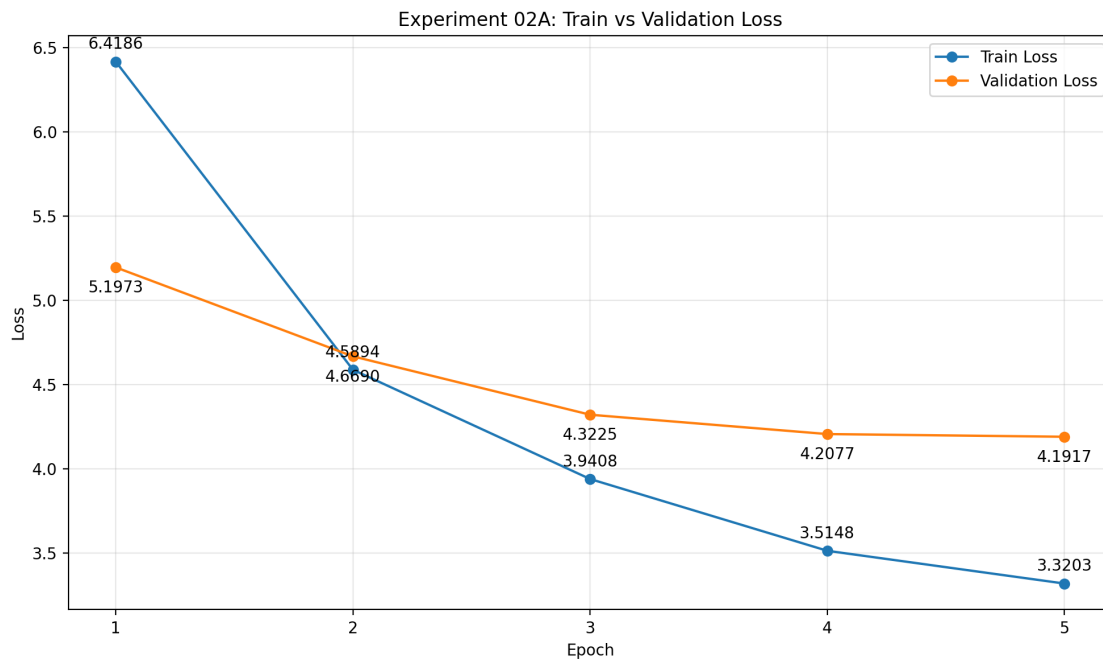


Figure 1. Training and validation loss across five epochs.

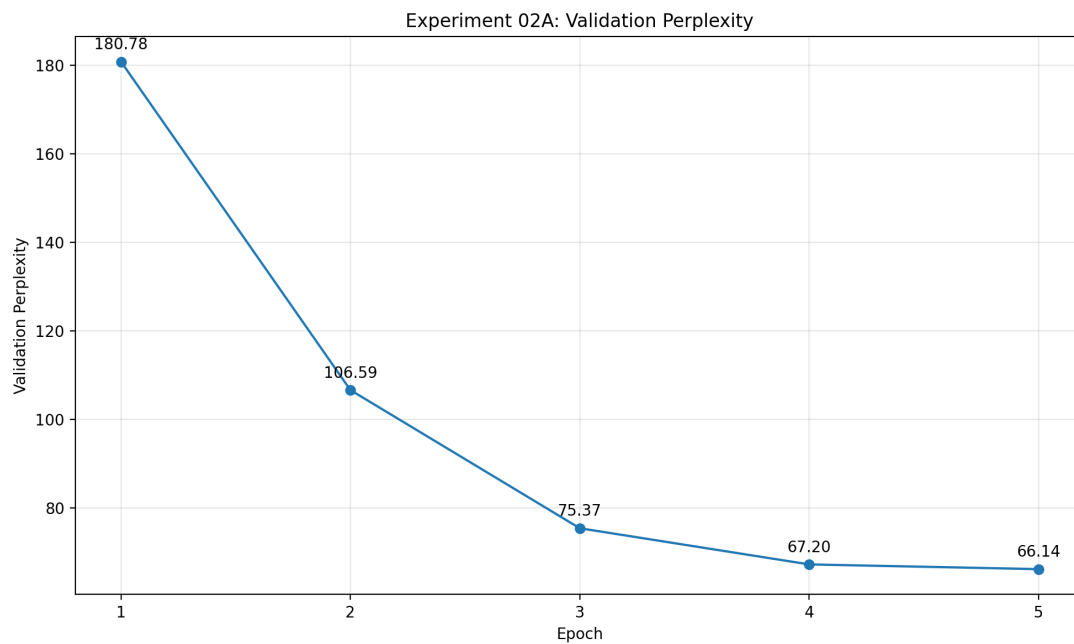


Figure 2. Validation perplexity improved steadily across training.

8. Learning Rate and Generalization Behavior

The learning rate followed the intended decay schedule and reached the minimum learning rate near the final epoch. The generalization gap stayed controlled relative to Experiment 01, indicating that the scaled setup reduced the overfitting failure mode observed in the initial baseline.

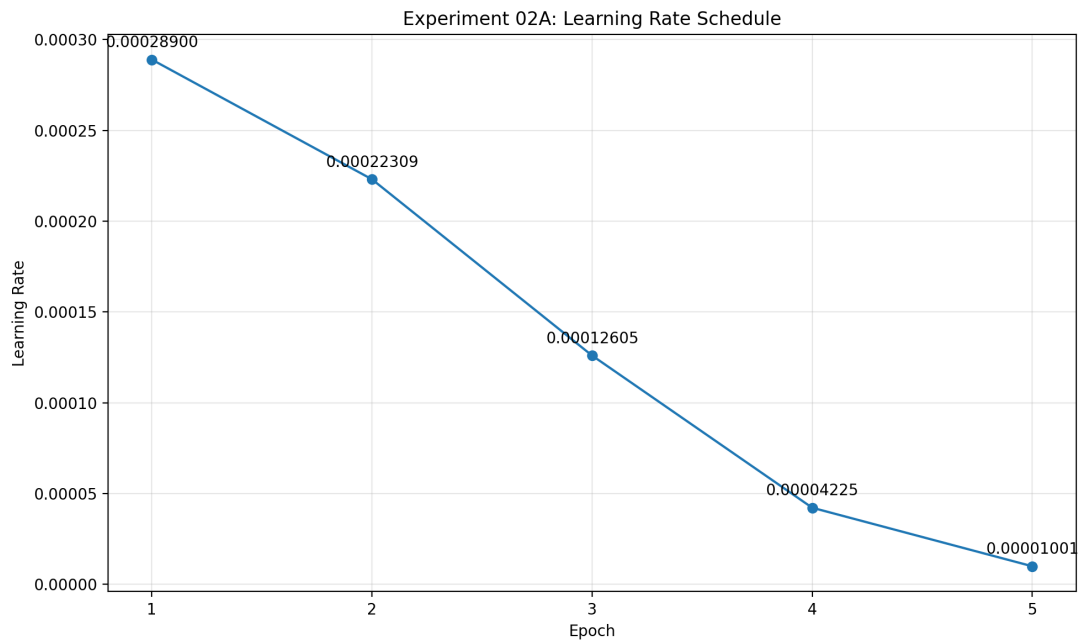


Figure 3. Learning-rate schedule across epochs.

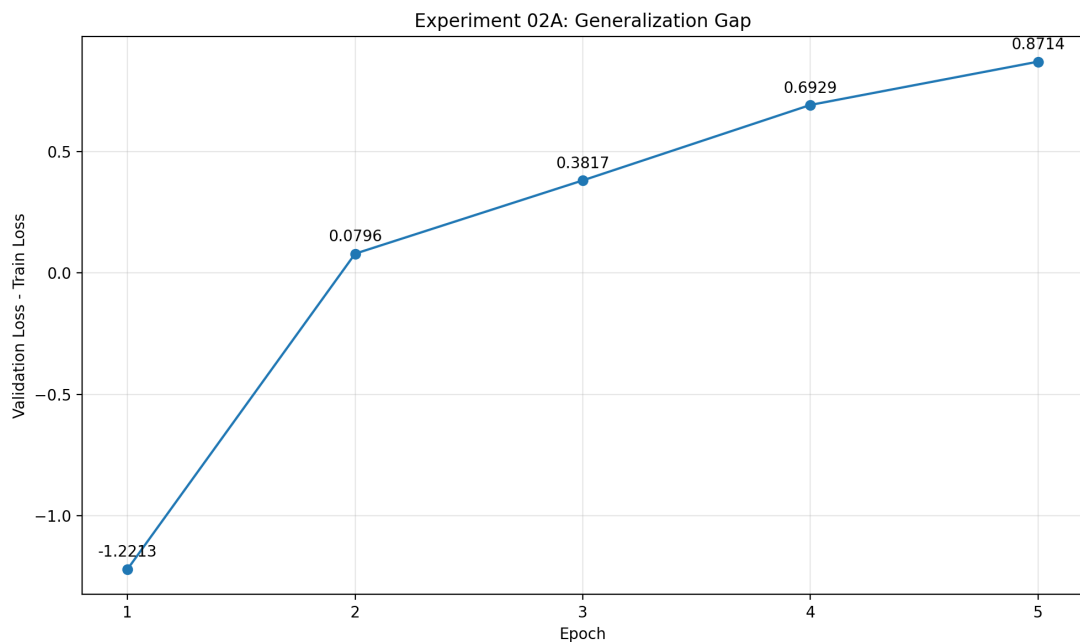


Figure 4. Generalization gap measured as validation loss minus training loss.

9. KV-Cache Benchmark

A KV-cache benchmark was used to compare autoregressive generation with and without cached key-value tensors. The benchmark evaluated prompt lengths of 64, 128, 256, and 512 tokens while generating 64 new tokens.

Prompt Length	No KV Latency	KV Latency	Speedup	No KV Tok/s	KV Tok/s	No KV Mem	KV Mem
64	0.333337	0.303964	1.0966x	192.00	210.56	811.95 MB	786.86 MB
128	0.334250	0.306064	1.0921x	191.50	209.13	836.68 MB	820.13 MB
256	0.328982	0.302917	1.0860x	194.55	211.29	886.98 MB	858.34 MB
512	0.456390	0.296575	1.5389x	140.23	215.80	985.32 MB	963.28 MB

The strongest benefit occurred at prompt length 512, where KV-cache reduced latency from 0.456390 seconds to 0.296575 seconds and improved throughput from 140.23 tokens/sec to 215.80 tokens/sec.

10. KV-Cache Plot Analysis

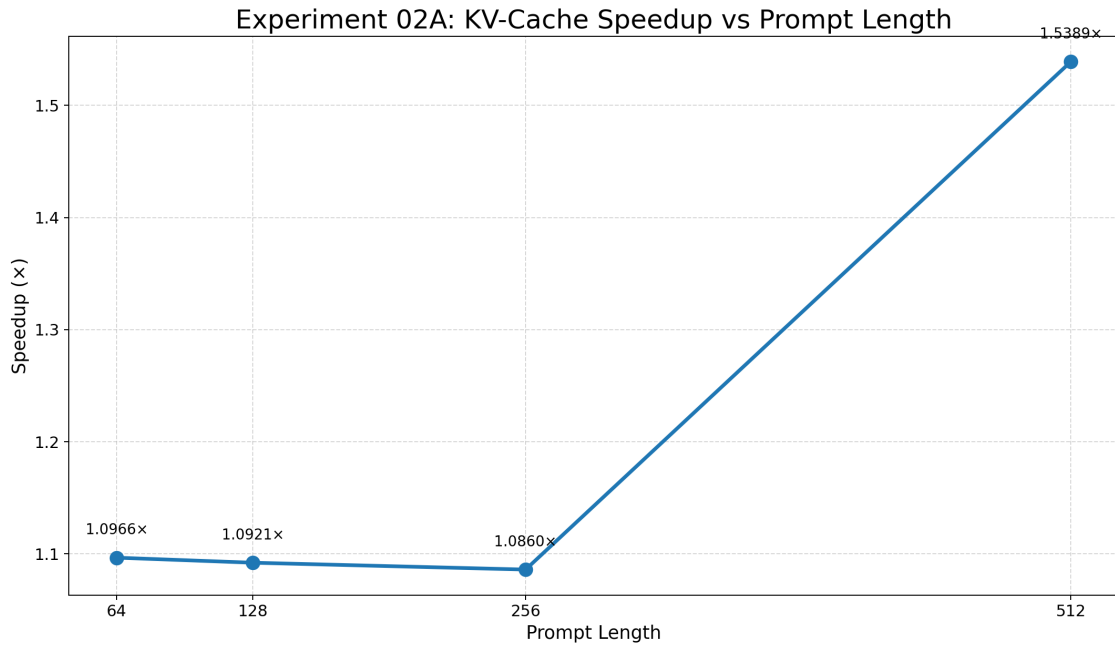


Figure 5. KV-cache speedup increased most clearly at the longest prompt length.

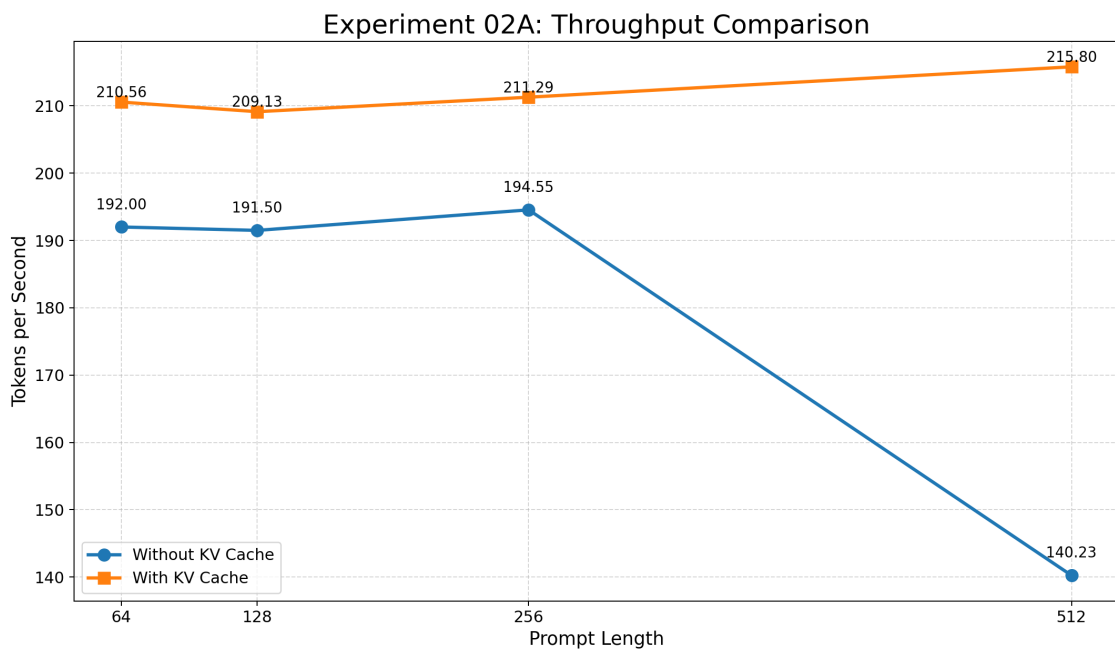


Figure 6. KV-cache improved generation throughput, especially at prompt length 512.

11. Latency and Memory Behavior

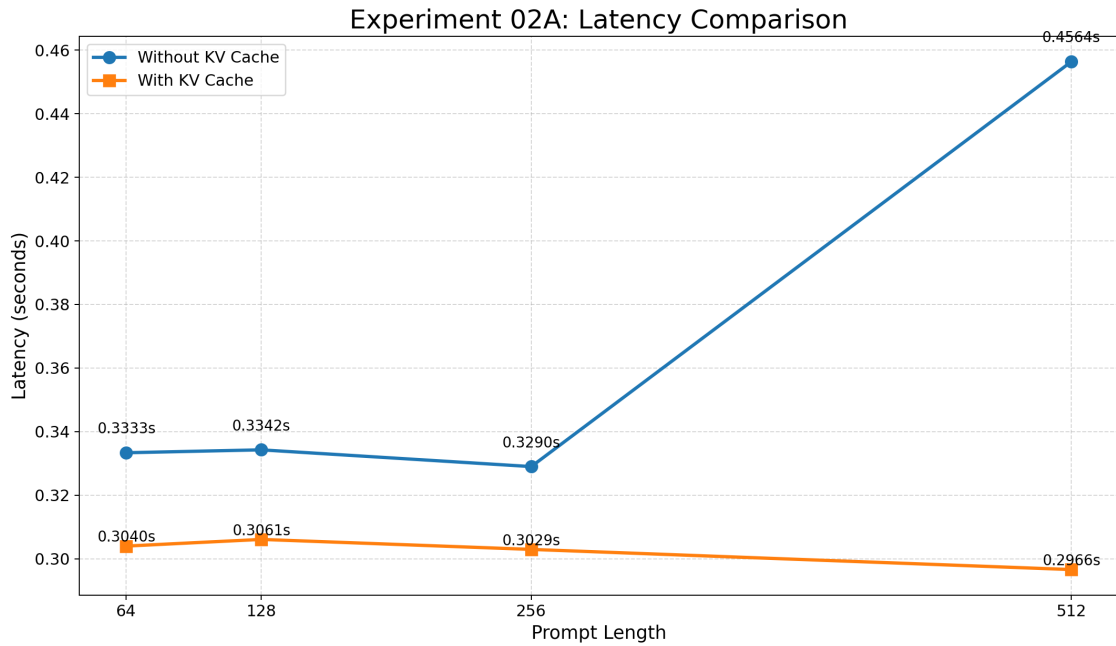


Figure 7. Latency comparison with and without KV-cache.

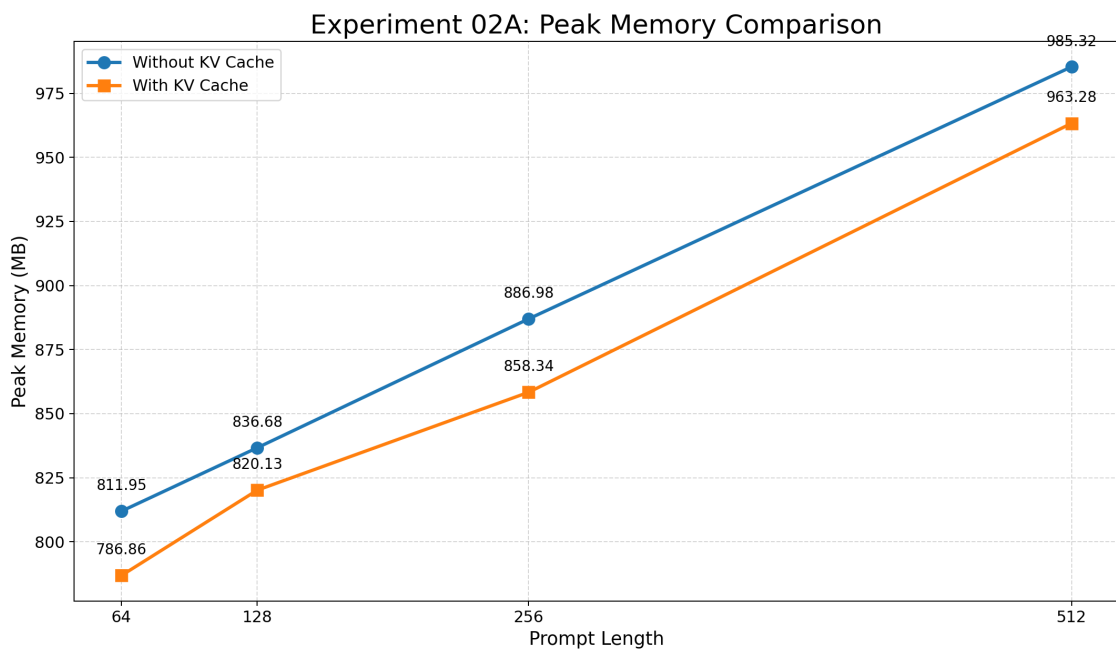


Figure 8. Peak memory comparison with and without KV-cache.

12. Comparison with Experiment 01

Experiment 02A successfully addressed the major limitations identified in Experiment 01. Instead of rapidly memorizing a small WikiText subset, the model showed stable validation improvement on a larger medical-domain setup.

Area	Experiment 01	Experiment 02A
Dataset	WikiText-103 subset	Medical text corpus
Context Length	128	1024
Model Scale	~16M parameters	~194M parameters
Checkpointing	latest.pt only	best.pt, latest.pt, epoch_N.pt
Validation Behavior	Validation loss increased	Validation loss decreased
KV-cache Benchmark	Not included	Included
Hardware	Google Colab T4	Lightning H200

13. Future Work: Experiment 02B

Experiment 02A established the scaled GPT-2/tiktoken baseline. The next planned experiment, Experiment 02B, replaces the generic GPT-2 tokenizer with a custom biomedical BPE tokenizer while keeping the standard attention setup.

The purpose of Experiment 02B is to evaluate whether a domain-specific tokenizer improves medical-domain language modeling quality compared with the generic GPT-2/tiktoken baseline.

Feature	Experiment 02A	Experiment 02B
Tokenizer	GPT-2 BPE / tiktoken	Custom biomedical BPE
Attention	Standard causal attention	Standard causal attention
Dataset Domain	Medical corpus	Medical corpus
Purpose	Scaled baseline	Tokenizer comparison

14. Conclusion

Experiment 02A successfully transformed the project from a small debugging baseline into a stronger, reproducible medical-domain GPT training experiment. The model achieved stable validation behavior, improved perplexity across epochs, reliable checkpoint preservation, and measurable KV-cache inference gains.

This experiment now serves as the official GPT-2/tiktoken baseline for later comparisons with the custom biomedical tokenizer and attention optimization variants.